

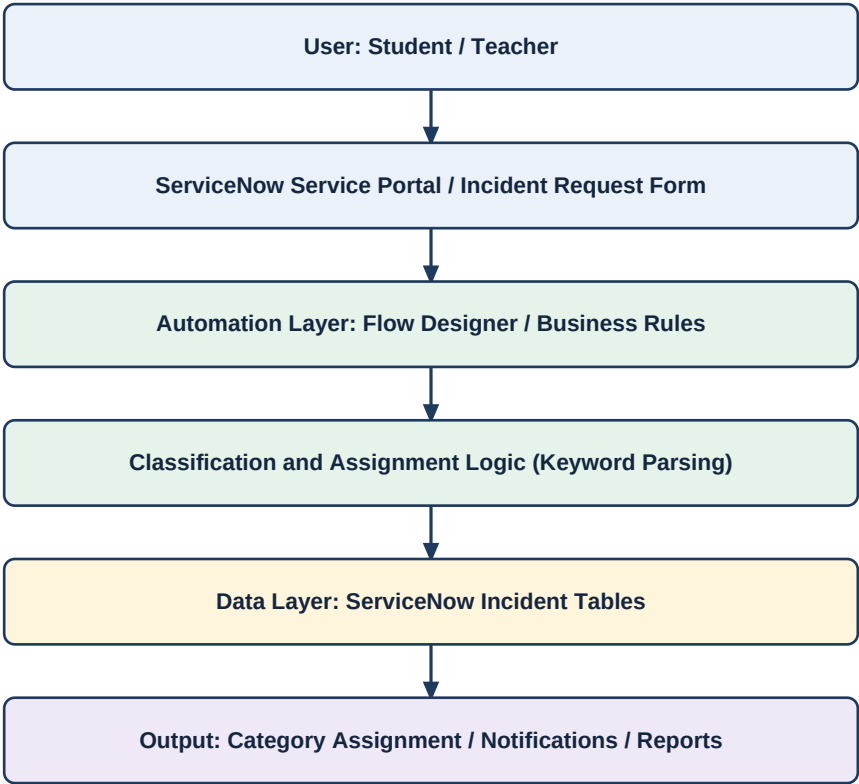
Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	
Project Name	Auto Ticket Classification using Flow Designer
Team Leader	AATHITHYA K V
Team Member	AATHITHYA K V
Team Size	1
Maximum Marks	4 Marks

Technical Architecture

Project Objective: Automate the ticket classification workflow in ServiceNow for the school IT helpdesk. Students and teachers submit incident requests (Wi-Fi issues, projector failures, password problems, slow computers). Flow Designer analyzes keywords in the short description and description fields, automatically assigns the correct category and IT assignment group, and sends status notifications with minimal manual effort.

Architecture Diagram Flow



Deployed Platform: The solution is deployed on the ServiceNow cloud platform.

ServiceNow Data Tables and Components

Layer	Components
User	Student / Teacher / Requester
ServiceNow Service Portal	Incident Request Form (Short Description and Description)
Automation Layer	Flow Designer, Business Rules, Script Includes
Approval and Routing	IT Support Assignment Groups
Data Layer	Incident [incident] table, Category table, Work Notes / Logs
Output	Automated Category, Email Notifications, IT Helpdesk Dashboard

Table 1: Technology Stack / Architecture Components

S.No.	Component	Description	Technology
1	User Interface	Student and teacher-facing incident reporting form and Helpdesk Portal.	ServiceNow Service Portal / Incident Form / Workspace
2	Application Logic-1	Workflow logic for keyword parsing, automatic category assignment, and task routing.	Flow Designer
3	Application Logic-2	Server-side logic for keyword pattern matching and automated field validation where required.	Business Rules, Script Includes, Glide APIs
4	Application Logic-3	Priority calculation and assignment group routing logic.	Assignment Rules, Flow Designer
5	Database	Stores incident tickets, categories, user details, assignment rules, and logs.	ServiceNow platform database / Application tables
6	Cloud Database	Managed cloud data layer provided by the ServiceNow platform.	ServiceNow Cloud Platform
7	File Storage	Optional storage for screenshot attachments or supporting error logs submitted by users.	ServiceNow Attachments / Platform file storage
8	External API-1	No mandatory third-party API required; integration can be added for external IT alerting.	ServiceNow REST API / IntegrationHub (Optional)
9	External API-2	Optional integration with school communication tools (e.g., Teams/Email notifications).	REST/SOAP APIs, IntegrationHub
10	Machine Learning Model	Optional Predictive Intelligence (Keyword/Text Analytics using Flow Designer logic).	ServiceNow Predictive Intelligence / Flow Designer
11	Infrastructure (Server/Cloud)	Application is hosted on the ServiceNow cloud platform; no local infrastructure required.	ServiceNow Cloud / SaaS Platform

Table 2: Application Characteristics

S.No.	Characteristics	Description	Technology
1	Open-Source Frameworks	Uses the native ServiceNow platform. JavaScript is used for server-side scripting and keyword matching logic.	JavaScript / ServiceNow APIs
2	Security Implementations	Role-based access control (RBAC), ACLs, authentication, and secure access to student/teacher data and IT incident records.	ServiceNow ACLs, Roles, Authentication, HTTPS/TLS
3	Scalable Architecture	Cloud-based SaaS architecture handles high volumes of daily incident tickets without maintaining local servers.	ServiceNow Cloud Platform, Flow Designer

S.No.	Characteristics	Description	Technology
4	Availability	Runs continuously on ServiceNow cloud services, enabling 24/7 access for students and teachers across the campus.	ServiceNow Cloud / SaaS
5	Performance	Instant classification occurs upon ticket creation using efficient Flow Designer triggers without delays.	Flow Designer, Business Rules, Glide APIs

Core Automated Flow

Step	Stage	Description
1	Incident Submission	Student/Teacher fills out the Incident form with Short Description and Description.
2	Trigger	Flow Designer triggers immediately upon new ticket creation.
3	Keyword Analysis	System scans Short Description and Description for target keywords (e.g., "Wi-Fi", "projector", "password", "slow").
4	Auto-Classification	Category, Subcategory, and Priority are assigned automatically.
5	Task Assignment	Incident is directly routed to the corresponding IT Support Group.
6	Notification	Requester receives automated notification with ticket number and assigned category.
7	Resolution and Closure	IT staff resolves the issue, updates resolution details, and closes the ticket.

Project Team Details

Role	Name	Team ID
Team Leader	AATHITHYA K V	
Team Member	AATHITHYA K V	

References

1. ServiceNow Documentation: Flow Designer, Incident Management, and Data Lookup Rules.
2. ServiceNow Developer Documentation: JavaScript APIs, GlideRecord, and Platform Architecture.